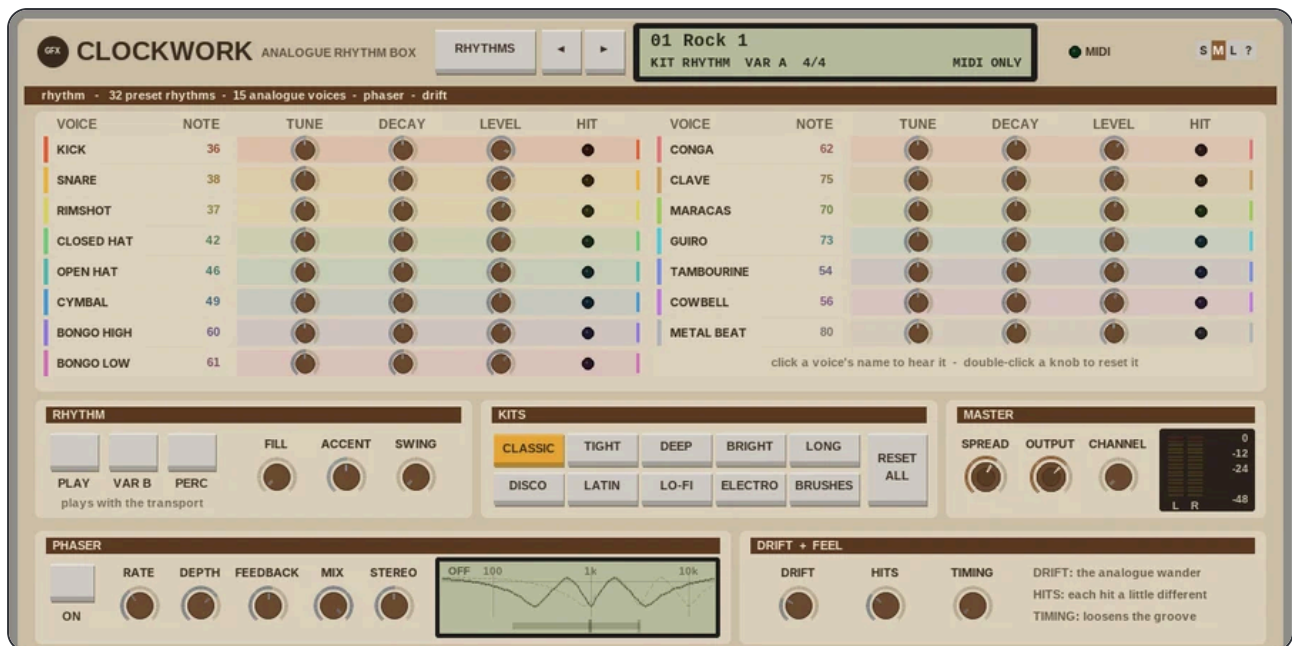


G Clockwork

An analogue preset-rhythm box: 15 synthesised voices, 32 built-in rhythms, ten kits and a phaser.



Overview

G Clockwork is a rhythm box in the spirit of the late-70s preset-rhythm machines. All 15 voices are synthesised, with no samples: kick, snare, rimshot, closed and open hat, cymbal, high and low bongo, conga, clave, maracas, guiro, tambourine, cowbell and metal beat. Each voice has its own TUNE, DECAY and LEVEL, and the KITS keys re-voice the whole machine in one go.

You can play it two ways, or both at once. From MIDI, each voice answers to its General MIDI drum note. Or switch on PLAY and it runs one of 32 built-in rhythms - 16 full-kit rhythms and 16 hand-percussion rhythms - locked to REAPER's tempo while the transport runs, with a B variation, automatic fills, accent and swing. A built-in phaser, the GFX instrument drift and per-hit variation keep it sounding like a real, slightly unruly machine.

Reach for it for

- Vintage drum-machine grooves for demos, songwriting and lo-fi tracks
- Hand-percussion loops to play along with (the PERC rhythms)

- Programming analogue-style beats from MIDI
- Swirling, phased drum loops
- Adding a shaker, tambourine or cowbell part under a live kit

Getting started

1. Insert a new track and add G Clockwork to it as an instrument. It makes its own sound and needs no audio input.
2. To play the built-in rhythms: click RHYTHMS (or the display) and pick a rhythm, switch PLAY on, then start REAPER's transport. The rhythm follows the project tempo and stops when the transport stops.
3. To play it from MIDI: record-arm the track with a MIDI input, or draw notes in a MIDI item on the track. Each voice answers to its General MIDI drum note (shown in the NOTE column). The MIDI light in the header flashes on each note.
4. Both work at the same time: leave PLAY on and add extra hits from MIDI on top.

Controls

The header

RHYTHMS	Opens the rhythm list, in two groups: KIT RHYTHMS and PERCUSSION ONLY.
◀ / ▶	Step to the previous or next rhythm.
Display	Shows the rhythm's number and name, whether it is a KIT RHYTHM or PERCUSSION, VAR A or VAR B, and its feel (4/4, 3/4 or TRIPLET). On the right: the current BAR (and FILL while a fill plays), WAIT while PLAY is on and the transport is stopped, or MIDI ONLY when PLAY is off. Click it to open the rhythm list too.
MIDI	Lights briefly on each incoming MIDI note.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.

?

Next to the S / M / L size keys. Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

The voices

The 15 voices sit in two columns. Click a voice's name to hear it. Double-click a knob to reset it.

VOICE	The voice's name tab. Click it to hear the voice.
NOTE	The MIDI note that plays the voice: KICK 36 (and 35), SNARE 38 (and 40), RIMSHOT 37, CLOSED HAT 42 (and 44), OPEN HAT 46, CYMBAL 49 (and 51, 52, 55, 57, 59), BONGO HIGH 60, BONGO LOW 61, CONGA 62 (and 63, 64), CLAVE 75, MARACAS 70, GUIRO 73 (and 74), TAMBOURINE 54, COWBELL 56, METAL BEAT 80 (and 81). Note velocity sets how hard each hit is.
TUNE	The voice's pitch, -12 to +12 semitones.
DECAY	How long the voice rings, 25 to 400% of its normal length.
LEVEL	The voice's level in the mix, 0 to 100%.
HIT	Lights in the voice's colour each time it plays.

The closed hat cuts off the open hat, as on the original machines. Each voice also has its own fixed place in the stereo picture, which SPREAD widens or narrows.

RHYTHM

PLAY	Plays the selected rhythm while REAPER's transport runs, locked to the project's tempo and bars. Off: the machine only plays from MIDI.
VAR B	Adds the variation parts on top of the basic rhythm (extra percussion, open hats, cymbals).
PERC	Switches between a kit rhythm and its percussion-only partner (rhythm 1 and 17, 2 and 18, and so on).

FILL	Drops a fill into the second half of the last bar: OFF, EVERY 2, EVERY 4 or EVERY 8 bars.
ACCENT	How much the accented hits stand out, 0 to 100% (50% by default). At 100% the unaccented hits play at about half strength.
SWING	Delays the off-beat 16ths, 0 to 100%, for a shuffled feel. Works on the 16-step rhythms only; the triplet and 3/4 rhythms are left as they are.

KITS

CLASSIC ... BRUSHES	Ten keys, each with its kit's name on it, that re-voice the whole kit (tune, decay and the balance between voices) - see Kits below. The lit key shows the last kit chosen. Your rhythm settings are left alone.
RESET ALL	The key beside the kits: every control back to its starting value.

MASTER

SPREAD	How wide the voices are spread across the stereo picture, 0 to 100% (60% by default). 0 puts everything in the centre.
OUTPUT	Output level, -24 to +12 dB.
CHANNEL	The MIDI channel the voices listen to: OMNI (all channels, the default) or channel 1 to 16. MIDI is always passed on down the chain.
Meter	Output level, left and right, with a peak hold.

PHASER

A six-stage phaser on the whole machine, swept by its own LFO. The display shows the phaser's response as it sweeps, with a marker below it for the sweep position.

While the phaser is off the display is dimmed and reads OFF. It fades in and out smoothly, so switching it on or off never clicks.

ON	Switches the phaser in or out.
RATE	The sweep speed, 0.05 to 8 Hz (0.3 Hz by default).
DEPTH	How far the sweep travels, 0 to 100% (70% by default).
FEEDBACK	Resonance around the phaser, 0 to 95% (50% by default). Higher settings give a sharper, more whistling sweep.
MIX	How much phaser, 0 to 100% (100% by default).
STEREO	Offsets the right side's sweep from the left, 0 to 180 degrees (90 by default). 0 is a mono phaser; higher settings give wide stereo movement.

DRIFT + FEEL

DRIFT	The GFX instrument drift, 0 to 100% (25% by default): the tuning of the whole kit wanders slowly (up to about 20 cents), the brightness of the noise voices (snare, hats, maracas, tambourine) and the tone of the metal voices (cymbal, cowbell, metal beat) drift, decays vary slightly, and each hit is tuned a little differently. 0 switches it off.
HITS	Makes every hit a little different in tune, decay and level, 0 to 100% (30% by default).
TIMING	Loosens the groove: each hit is delayed by a random amount up to this time, 0 to 20 ms (0 by default). Works on rhythm and MIDI hits alike.

IN / OUT strip

SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
------------	---

OFF / CLIP / LIMIT (OUT)	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default; its key flashes red while it is clipping.
MONO	Makes the low end mono below this frequency, from OFF up to 500 Hz.
WIDTH	Widens the stereo picture, from +0% (the default) to +100%.

Presets

Kits

Each kit first sets every voice back to its normal level, then changes the tune, decay and balance.

CLASSIC	The machine as designed: every voice at its normal tune and decay.
TIGHT	Tight & Dry: every voice shortened to 55% (the kick to 70%).
DEEP	Deep & Boomy: tuned down 3 semitones with longer decays; the kick 4 semitones down with twice its decay.
BRIGHT	Tuned up 3 semitones (the kick 2), slightly shorter.
LONG	Long Tails: every voice rings much longer (220%, the kick 160%).
DISCO	A little higher and tighter, with a deep, long kick and louder kick, hats and tambourine.
LATIN	Latin Percussion: kick and snare down, bongos, conga, clave, maracas, guiro and cowbell up.
LO-FI	Lo-Fi Box: tuned down and shorter, with quieter hats and cymbal.

ELECTRO	Short and slightly low, with a very deep, long kick, a punchy short snare, and louder kick and snare.
BRUSHES	Jazz Brushes: a long, low, soft snare, a quieter kit, longer decays and a shorter kick.

Kit rhythms

All kit rhythms have a kick-and-snare fill.

01 Rock 1	Kick, snare on 2 and 4, eighth-note hats. VAR B: open hat and a cymbal on the one.
02 Rock 2	Busier kick, sixteenth hats with accents. VAR B: tambourine and open hat.
03 Disco	Four-on-the-floor kick, snare on 2 and 4, open hats on the off-beats. VAR B: sixteenth tambourine and metal beat.
04 Funk	Syncopated kick and snare, accented sixteenth hats. VAR B: open hat and cowbell.
05 Ballad	Gentle kick, rimshot on 2 and 4, eighth hats. VAR B: sixteenth maracas.
06 Pop	Kick, snare, accented eighth hats. VAR B: tambourine backbeat and a cymbal.
07 Shuffle	A triplet shuffle on kick, snare and hats. VAR B: open hat and cymbal.
08 Slow Rock	A 12/8 triplet rock beat. VAR B: cymbal and tambourine.
09 Swing	A triplet ride pattern on the cymbal, hats on 2 and 4, light kick. VAR B: rimshot.
10 Waltz	3/4: kick on the one, rimshot on two and three, hats. VAR B: cymbal and maracas.
11 Bossa Nova	Kick, rimshot clave pattern, sixteenth maracas. VAR B: closed hat and conga.

12 Samba	Kick, sixteenth tambourine, high bongo. VAR B: conga and cowbell.
13 Rhumba	Clave, kick, low bongo, conga, maracas. VAR B: high bongo and guiro.
14 Cha-Cha	Kick, cowbell on the beat, guiro, conga. VAR B: clave and high bongo.
15 Tango	Kick on every beat, rimshot, hats. VAR B: conga and tambourine.
16 Beguine	Kick, bongos and maracas. VAR B: cymbal and clave.

Percussion rhythms

No kick, snare, hats or cymbal: hand percussion to play along with. Each has its own percussion fill.

17 Tumbao	Conga, clave and maracas. VAR B: bongos.
18 Salsa	Clave, cowbell, conga and guiro. VAR B: bongos.
19 Bossa Perc	Rimshot, maracas and conga. VAR B: bongos.
20 Batucada	Tambourine, cowbell, metal beat, conga and high bongo. VAR B: maracas and low bongo.
21 Rumba Perc	Clave, maracas, conga and low bongo. VAR B: guiro and high bongo.
22 Mambo	Cowbell, clave, conga and guiro. VAR B: metal beat and high bongo.
23 Afro 6/8	A triplet bell pattern on cowbell, with maracas, conga and low bongo. VAR B: high bongo and tambourine.
24 Calypso	Tambourine, cowbell, conga and low bongo. VAR B: high bongo and maracas.
25 Reggae Perc	Rimshot, maracas, low and high bongo. VAR B: tambourine

	and clave.
26 Merengue	Guiro, conga and cowbell. VAR B: high bongo and maracas.
27 Baiao	Metal beat, conga and low bongo. VAR B: maracas and high bongo.
28 Shaker Pop	Accented maracas, tambourine and conga. VAR B: high bongo and clave.
29 Tamb Soul	Tambourine, clave and conga. VAR B: maracas and low bongo.
30 Martillo	High and low bongo with cowbell. VAR B: conga and clave.
31 Guaguanco	Clave, conga and rimshot. VAR B: high bongo and maracas.
32 Cumbia	Guiro, maracas, conga and low bongo. VAR B: high bongo and tambourine.

Tips

- Pick a kit rhythm, then press PERC to hear its percussion partner - great for adding a shaker or conga part under a live drummer.
- Turn FILL to EVERY 4 or EVERY 8 so long loops don't sound stuck.
- A little SWING (20-40%) and a few ms of TIMING make the 16-step rhythms feel much less mechanical.
- For a classic swirling drum-machine sound, switch the PHASER on with a slow RATE and FEEDBACK around 50-70%.
- Use the MIDI notes to add your own hits on top of a running rhythm - a crash on the one, an extra conga, a cowbell part.

File: gfx-g-clockwork.jsfx